

99 per cent from up to eight metres away. The system as a whole can track more than one occupant, at all times, whichever room he or she might be in, whether sitting, sleeping or moving.

Researchers came up with this device based on the fact that wireless signals are impacted by even the smallest motion in the environment, even something as mild as our chest heaving during inhalation or exhalation and vibrations caused on our skin when the heart beats.

The monitor first identifies the person's presence, focuses on him or her and then measures the heart rate and breathing pattern by transmitting a low-power wireless signal and measuring the time it takes for the signal to be reflected back to the device. The patterns are then analysed by the device.

The key benefit of this device is that it is totally non-intrusive. You do not need to wear anything at all. So the device can constantly watch the occupants of a house and, over time, apps can be used to study patterns, identify increase in stress levels, abnormal health conditions and so on. Now, that sounds even more useful than energy saving, right?

Love them abundantly, but watch only silently

A real-life story covered by Heather Kelly in CNN's Tomorrow Transformed shows how two daughters used a highly-customisable, do-it-yourself (DIY) home automation system from SmartThings to unobtrusively keep an eye on their 77-year-old mom, Mary Lou, who was in the middle stages of Alzheimer's.

When Lou leaves home at an odd hour, a silent sensor on her front door alerts her daughters on their respective phones. Sensors on her keychains help find out if she leaves her condo premises, while some in the kitchen keep an eye on her eating habits. More sensors in the bedroom track when she sleeps, how she sleeps and when she wakes up, while flood sensors in the laundry room and smoke sensors in the kitchen assure her daughters that Lou is safe.

The sensors are connected to a local wireless hub and together with a smartapp, these help Lou's daughters give her the satisfaction of living independently and safely in her own home, while they remain assured of her safety.

There are other companies like Lively, BeClose and GrandCare that offer connected accessories such as beds, toilets, pillbox sensors and blood-pressure, weight- and glucose-monitoring devices. Most of these work silently without invading senior citizens' privacy, which is most important to let them live with dignity.

Tech gives the gift of independence

Not just senior citizens, there are also many youngsters with disabilities who are beginning to use technology to

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